

The empirical recurrence for OEIS A301441, recovered from its tail

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Abstract

OEIS A301441 records “Empirical recurrence of order 97” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 227$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 99$. Its last coefficient is $c_{97} = 32148900 \neq 0$, so no further tail satisfies anything shorter; any order-97 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A301441 is “Number of $n \times 6$ 0..1 arrays with every element equal to 0, 1, 2, 3, 4 or 6 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

32, 2048, 92616, 4543028, 221935656, 10857946336, 531234388944, 25991845818768, ...

The entry states:

Empirical recurrence of order 97 (see link above)

As of the “Last modified” line on the live entry (Mar 21 2018 11:46:31, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 227$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 227$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 97: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 97.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 454$ exact terms from $n = 2$ on, it returns order 97 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{97} c_i a(n-i),$$

c_1	64	c_{26}	7170515039214577	c_{51}	−248595254038217454671	c_{76}	732913221577
c_2	−766	c_{27}	14364454762437046	c_{52}	273828012644180302901	c_{77}	1950692367726
c_3	999	c_{28}	−35420820397445929	c_{53}	−500399762752157931957	c_{78}	746462393734
c_4	19981	c_{29}	−65136768491398788	c_{54}	2858480434954970890886	c_{79}	135406747632
c_5	−17084	c_{30}	−74370256206457398	c_{55}	−662504389723555804736	c_{80}	−84733545693
c_6	−471143	c_{31}	259558014034674421	c_{56}	−1989817106699109752063	c_{81}	−153289503726
c_7	335886	c_{32}	−55097801349219607	c_{57}	1040892150260631579521	c_{82}	−63614730270
c_8	5040267	c_{33}	−227466950898724441	c_{58}	−2901045282170402310395	c_{83}	89373359846
c_9	6628014	c_{34}	−943342221193099178	c_{59}	180468974327948539802	c_{84}	146767745757
c_{10}	−56697945	c_{35}	3297799273608126210	c_{60}	2962616200164481536536	c_{85}	39539595248
c_{11}	−187441043	c_{36}	3988140413863094738	c_{61}	170786481831639096958	c_{86}	−2657922006
c_{12}	396173992	c_{37}	2623968746140063056	c_{62}	1072366323329493268198	c_{87}	−4479847127
c_{13}	1837530956	c_{38}	−9868859393275651821	c_{63}	184659589043508000414	c_{88}	−1144320415
c_{14}	−2642154086	c_{39}	−6725075126497691546	c_{64}	−1537979590718410873127	c_{89}	1878563183
c_{15}	−10468783534	c_{40}	334046566944907917	c_{65}	−528554205307891378266	c_{90}	6310141784
c_{16}	98149056723	c_{41}	−19042852901824975157	c_{66}	−265146528483648301737	c_{91}	−167107626
c_{17}	161519215882	c_{42}	−28306953703157600623	c_{67}	−105068033171777146333	c_{92}	−481641380
c_{18}	−1065866819290	c_{43}	−42917205869972088922	c_{68}	332572562041595050826	c_{93}	216004149
c_{19}	−3709308349488	c_{44}	25597089559154110503	c_{69}	164558901353316043650	c_{94}	276433935
c_{20}	651631474775	c_{45}	54717082940739202634	c_{70}	31823984276038172337	c_{95}	−6684995
c_{21}	61251878690736	c_{46}	107782329088373959410	c_{71}	22609939892760755441	c_{96}	−5393645
c_{22}	17569797592963	c_{47}	508493423938698942651	c_{72}	−29139371327657657137	c_{97}	32148900
c_{23}	−276640052235544	c_{48}	−114695672318302134335	c_{73}	−21124002733051236562		
c_{24}	−1329272268048324	c_{49}	−171066939162559044905	c_{74}	−3363436278552013019		
c_{25}	1647124495895316	c_{50}	−816792796783941833403	c_{75}	−2515074708424951512		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 99$, and it is the recurrence OEIS A301441 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{97} - c_1x^{96} - \dots - c_{97}$. Its constant term is $-c_{97} = -32148900$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 97 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 97, so $\deg q = 97$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 97, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 14 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 97, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 32148900.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-97 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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